AI101 COURSE PROJECT

Predicting a City's Water Consumption using Machine Learning

Team 62

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Repository: **github.com/Paramjitsingh7/AI-mini-project-group-62**

The Problem and Data

What and why: A machine-learning mini-project analyzing historical monthly municipal water consumption. We use Linear Regression to predict consumption for August 2009 based on the preceding three months.

178,597

rows × 89 columns

ORIGINAL SIZE

141,205

rows × 89 columns

CLEANED SIZE

DATASET SOURCE kaggle.com/datasets/marcomolina/water-consumption-in-a-median-size-city

Approach

1 → 2 → 3 → 4 → 5

Data Preprocessing & Cleaning

Training and Testing Data Split

Model Training Linear Regression

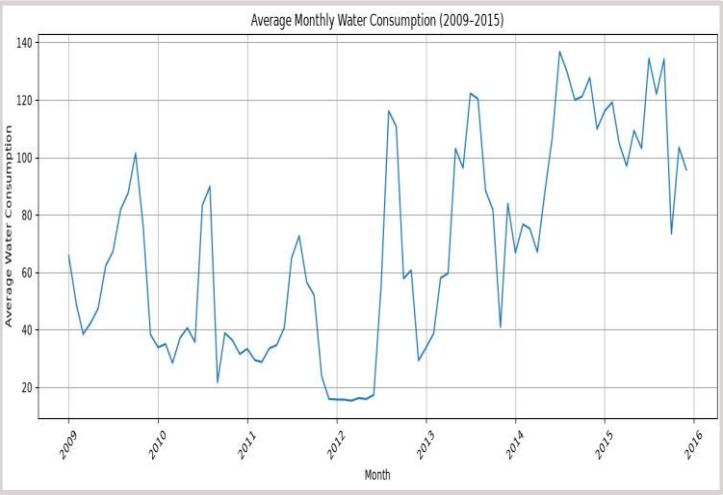
Water Consumption Prediction

Model Performance Evaluation

LIBRARIES USED

pandas · numpy · matplotlib · scikit-learn

Results



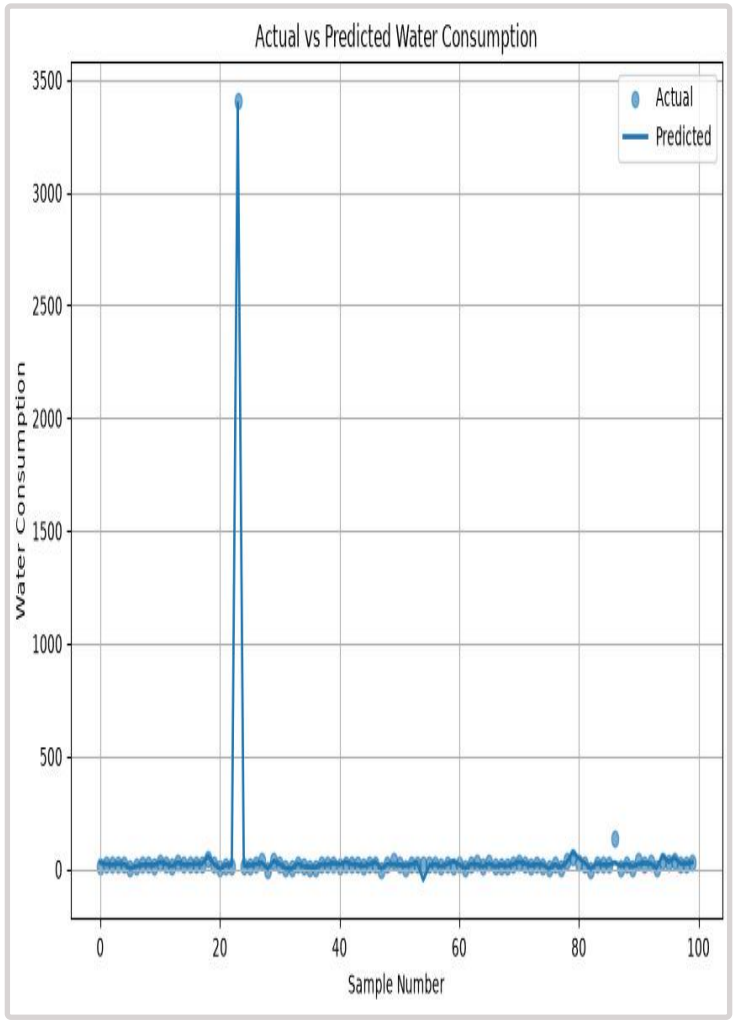
3.Predicting the AI Model

In [ ]:
```
y_pred = model.predict(X_test)
```

In [ ]:
```
results = pd.DataFrame({
    "Actual": y_test,
    "Predicted": y_pred
})

display(results.head())
```

	Actual	Predicted
96887	11.0	28.264698
163807	20.0	21.010423
12812	21.0	18.711256
55183	25.0	23.047493
135816	20.0	21.010423



R^2 Score

0.9991

MAE

5.1660

RMSE

14.6159

MSE

213.6243

Learnings & Disclosure

WHAT WORKED

Effective data cleaning and preprocessing.
High model performance ($R^2 = 0.9991$).
Predictions closely matched actual values.

WHAT COULD BE IMPROVED

Limited input to only three months.
Prediction focused only on August 2009.

WHO DID WHAT

Param: cleaning and understanding data, execution
Sarthak: training the AI model, documentation
Ankit: predicting with the AI model, video
Banoth: error analysis, presentation

AI USE DISCLOSURE

AI tools were used during parts of the project workflow. ChatGPT was used by team members for understanding the dataset and its structure, clarifying Git and GitHub concepts and usage, improving the formatting and structure of the project presentation, and drafting the initial README.md file.

The team reviewed the AI-generated suggestions and used them as guidance throughout the project. AI assistance was primarily used to support learning, improve workflow efficiency, and enhance project documentation, while the team remained responsible for the final decisions and submitted work.